

教育经历

2022.09 - 2026.07	浙江大学	计算机科学与技术	博士	
2019.09 - 2022.07	南方科技大学	生物医学工程系	硕士	Rank:5%
2015.09 - 2019.07	长春理工大学	光电工程学院（国际精英班）	本科	Rank:20%

项目经历

2024.12 - 2025.08	基于材质和光照分离的逆问题重建	项目负责人
	- 项目1：设计压缩光谱重建算法，解耦光谱材质和灰度光照 - 项目2：建立RGB图片本征图像分解理论，使用逆问题迭代优化分离反照率和阴影 - 项目3：设计多视角三维重建算法，利用高斯泼溅重建几何、材质和光照 - 项目产出：投稿一作计算机顶会 NeurIPS [1], AAAI [2], 发表 ICME [3]	
2022.06 - 2025.08	高通量智能成像理论与技术在工业检测中的应用（300万）	项目负责人
	- 项目内容：利用光谱压缩OCT系统实现工业样本的单次曝光三维重建 - 项目路线：搭建压缩光谱OCT硬件系统搭建，采集三维实验数据。Blender构建三维层析数据集，设计即插即用（PnP）三维重建算法，部署在GPU上 - 项目产出：搭建系统样机一套，发表一作光学期刊 Light: Science & Applications [16] 和国际光学会议 Imaging Systems and Applications[21], 投稿 Advanced Photonics[19]	
2023.07 - 2024.07	计算成像赋能激光智能焊接	项目负责人
	- 项目内容：利用视频编码压缩方法实现超快激光加工的实时监测 - 项目路线：搭建飞秒激光超快加工成像系统，采集超快加工实验数据。部署相机-DMD硬件同步，设计端到端的视频压缩编码成像算法重建 - 项目产出：发表一作光学期刊 Light: Advanced Manufacturing [15] 和国际光学会议 Imaging Systems and Applications [20]	
2020.07 - 2022.03	活体深层脑组织近红外二区超分辨成像研究	模块负责人
	- 项目内容：基于近红外二区聚合物量子点的生物荧光超分辨成像研究 - 项目产出：发表一作光学期刊 Optics and Lasers in Engineering [17], 化学期刊 Chemical Communications [18]	
其他主导一作文章包含扩散模型 [4] 和 数学逆问题重建 [5], 参与 NeRF/Gaussian Splatting [7-9]、高光谱重建 [10-12], 全息重建 [14, 22]		

科研成果

视觉算法领域（三维重建，光谱重建，逆问题）
[1] Xiaodong Wang, Zijun He, Ping Wang, Lishun Wang, Yanan Hu, Xin Yuan*. Spectral Compressive Imaging via Chromaticity-Intensity Decomposition. NeurIPS 2025. (CCF-A, 审稿中)
[2] Xiaodong Wang, Ping Wang, Zijun He, Mengjie Qin, Xin Yuan*. Progressive Flow-inspired Unfolding for Spectral Compressive Imaging. AAAI 2025. (CCF-A, 投稿中)
[3] Xiaodong Wang, Zijun He, Xin Yuan*. Texture-aware Intrinsic Image Decomposition with Model- and Learning-based Priors. ICME 2025. (CCF-B, 接收)
[4] Xiaodong Wang, Ping Wang, Zhangyuan Li, Xin Yuan*. Plug-and-play Diffusion Models for Image Compressive Sensing with Data Consistency Projection. ICIP 2025. (CCF-C, 接收, 计算摄影领域重要会议)
[5] Xiaodong Wang. User Manual for Model-based Imaging Inverse Problem. arXiv. 2025. (数学成像逆问题手册)
[6] Yanan Hu, Xiaodong Wang, Zijun He, Xin Yuan*. Lighten-MST: Low-light Spectral Reconstruction via Illumination-guided Spectral-aware Transformer. ICCV Workshop. 2025. (CCF-A, 接收)
[7] Yunhao Li, Xiaodong Wang, Ping Wang, Xin Yuan, Peidong Liu*. SCINeRF: Neural Radiance Fields from A Snaps

hot Compressive Image. **CVPR** 2024. (CCF-A, **Highlight**)

[8] Xiaoyue Li, Yunhao Li, **Xiaodong Wang**, Xin Yuan, Mark D Butala*, Gaoang Wang*. SCI-Gaussian: Optimizing 3D Gaussian Radiance Fields from a Snapshot Compressive Image. **ICASSP** 2025. (CCF-B)

[9] Yunhao Li, Xiang Liu, **Xiaodong Wang**, Xin Yuan*, Peidong Liu*. Learning Radiance Fields from a Single Snapshot Compressive Image. **arXiv**. 2024.

[10] Yurong Chen, Yaonan Wang, **Xiaodong Wang**, Xin Yuan, Hui Zhang*. LCTC: Lightweight Convolutional Thresholding Sparse Coding Network Prior for Compressive Hyperspectral Imaging. **IEEE TIP** 2025. (中科院1区Top, IF=13.7)

[11] Yifan Si, Zijian Lin, **Xiaodong Wang**, Sailing He*. A New Hyperspectral Reconstruction Method with Conditional Diffusion Model for Snapshot Spectral Compressive Imaging. **IEEE Trans. Instrum. Meas.** 2025. (中科院2区, IF=5.9)

[12] Yifan Si, Shuo Li, **Xiaodong Wang**, Sailing He*. ASP-Model: An Advanced Deep Learning Framework to Reconstruct Hyperspectral Cubes for Computed Tomography Imaging System. **IEEE Trans. Instrum. Meas.** 2025. (中科院2区, IF=5.9)

光学成像领域（工业三维，荧光显微）

[13] **Xiaodong Wang**, Xin Yuan*. In-situ Real-time Monitoring of Ultrafast Laser Processing Using Wide-field High-resolution Snapshot Compressive Microscopy. **Light: Advanced Manufacturing**. 2024. (JCR Q1, IF=10.6)

[14] **Xiaodong Wang**, Xin Yuan*, Liping Shi*. Optical Coherence Tomography—In situ and High-speed 3D Imaging for Laser Materials Processing. **Light: Science & Applications**. 2022. (中科院1区Top, IF=23.4)

[15] **Xiaodong Wang**, Dayan Li*. Spectra-separated Depth-of-field Extended Fluorescence Imaging Through Scattering Media Using Speckle Deconvolution. **Optics and Lasers in Engineering**. 2023. (中科院2区, IF=3.7)

[16] Zihan Yao#, **Xiaodong Wang**#, Changfeng Wu*, Xiaofeng Fang*. Photoswitchable Semiconducting Polymer Dots for Pattern Encoding and Superresolution Imaging. **Chemical Communications**. 2023. (中科院2区, IF=4.2)

[17] **Xiaodong Wang**, Mu Qiao*, Xin Yuan*. Physics-inspired Deep Fourier-domain Denoiser Empowered Dual-camera Compressive Coherence Tomography. **Advanced Photonics**. 2025. (中科院1区, IF=18.8, 投稿中)

[18] **Xiaodong Wang**, Miao Cao, Liping Shi, Jiao Geng, Xin Yuan*. Wide-field, High-resolution Snapshot Compressive Microscopy for Laser Materials Manufacturing. **Imaging Systems and Applications**. 2023. (光学会议)

[19] Gang Qu, **Xiaodong Wang**, Xin Yuan*. Plug-and-Play Deep Image Prior for Snapshot Optical Coherence Tomography. **Imaging Systems and Applications**. 2023. (光学会议)

[20] Yufei Dou, Miao Cao, **Xiaodong Wang**, Xing Liu, Xin Yuan*. Coded Aperture Temporal Compressive Digital Holographic Microscopy. **Optics Letters**. 2023. (中科院2区, IF=3.3)

[21] Zhihe Liu, Jie Liu, **Xiaodong Wang**, Feixue Mi, Dan Wang, Changfeng Wu*. Fluorescent Bioconjugates for Super-resolution Optical Nanoscopy. **Bioconjugate Chemistry**. 2020. (中科院2区, IF=3.9)

专业技能

- 熟练使用 Python/Matlab/Linux
- 熟悉 Solidworks/Zemax/FDTD/Labview
- 英语（雅思6.5）/粤语

学术服务

- 《生物光学实验》，《医学图像处理》，《计算成像》课程助教
- 学术会议 ICCV, ICIP 以及期刊 TCI, COL, RSL 审稿